

# Treatymaking and Indigenous Agriculture in North America

*Exploring agricultural outcomes across Numbered Treaty boundaries  
in early 20th-century Canada*

**Faculty Supervisor**

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**Research Assistants**

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## Project Overview

**Faculty**  
Kulsoom Hisam

**Institution**  
St. Olaf College

**Focus Region**  
Alberta, Saskatchewan,  
Manitoba — Canada

**Time Period**  
Early 20th century  
(1870s–1930s)

## What is this research about?

This project examines agricultural development on Indigenous reserves in Canada during the early 20th century. It focuses on the Numbered Treaties signed mainly in the 1870s, which carried varying agricultural provisions — promising different combinations of farm implements, cattle, and seed grain.

### The Numbered Treaties

Land-cession treaties signed 1871–1921 between the Crown and First Nations. Agricultural provisions varied: some promised plows, cattle, and seed; others offered less.

### Dept. of Indian Affairs

Federal body that regulated land use, farm inputs, and trade on reserves — directly shaping how much Indigenous farmers could produce and sell.

### The Wheat Boom

Canada's prairie wheat boom (1890s–1920s) transformed the regional economy. Did Indigenous farmers share in these gains, or face systematically different outcomes?

## 03 | Research Questions

Q1

**Did variation in treaty clauses affect agricultural productivity on Indigenous reserves?**

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The Numbered Treaties promised different combinations of agricultural inputs — plows, cattle, seed grain, and instruction. If these provisions actually shaped what farmers could grow, we should observe productivity differences that align with treaty type, not geography.

Q2

**During Canada's wheat boom, did Indigenous farmers share in the gains — or face systematically different outcomes?**

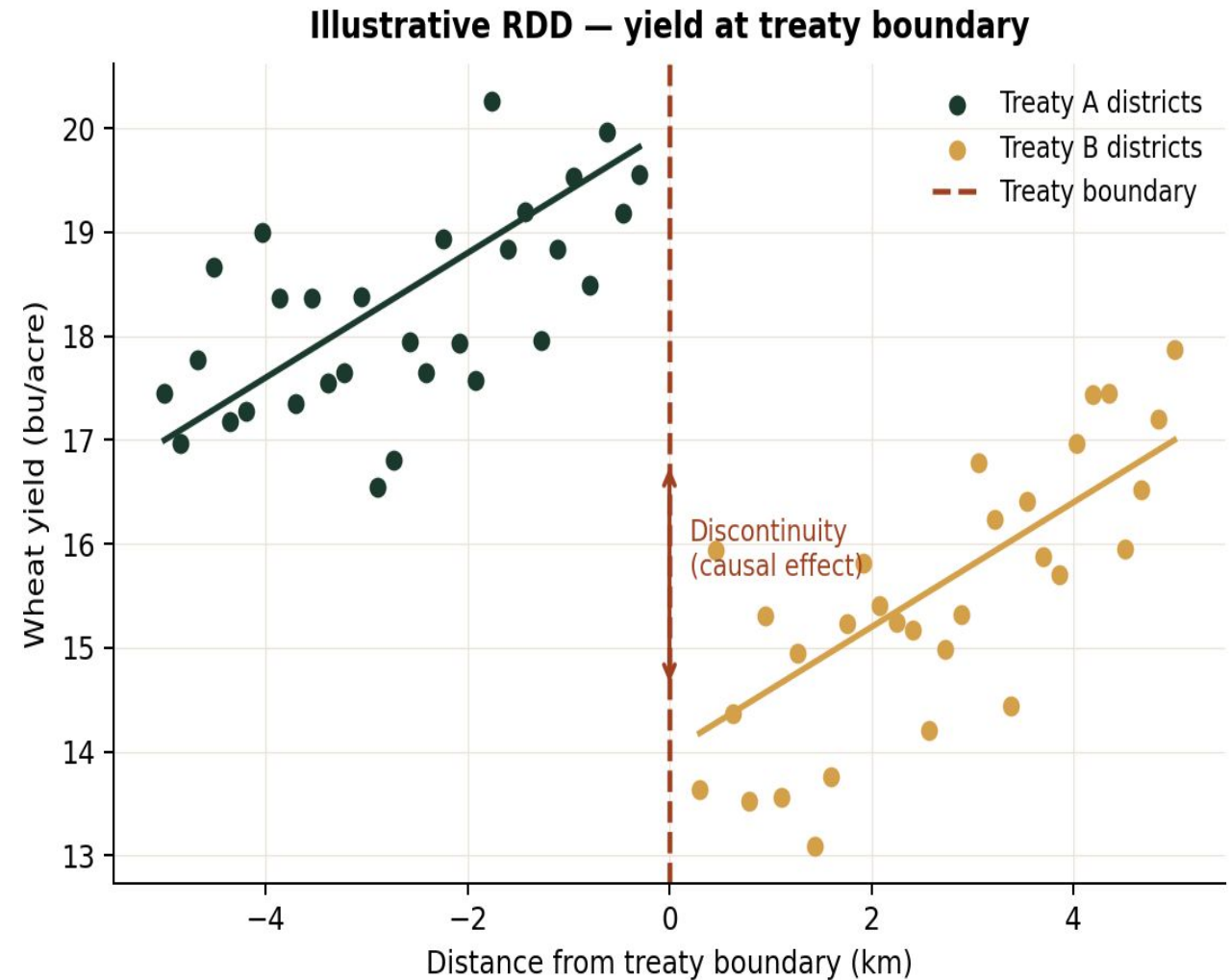
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The prairie wheat boom transformed the regional economy. Using district-level Annual Report data, we can test whether yields and acreage on reserves tracked the provincial trend, or diverged due to regulatory constraints imposed by the DIA.

*Both questions are addressed through a spatial regression discontinuity design comparing communities across treaty boundaries that are otherwise similar in geography and ecology.*

### Spatial Regression Discontinuity Design

- ▶ Communities near treaty boundaries are compared — they share similar geography, soil quality, and climate.
- ▶ The treaty boundary acts as a sharp cutoff: communities just across it were assigned different agricultural provisions.
- ▶ Any productivity differences at the boundary are attributed to treaty variation, not geography.
- ▶ This 'natural experiment' approach controls for many confounds that affect agricultural output.



## 05 | Data Sources

### SOURCE 01

#### DIA Annual Reports (Digitized PDFs)

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Annual Reports of the Department of Indian Affairs, published yearly and now digitized. They contain district-level agricultural statistics: acreage, total yield, and yield per acre for wheat, oats, barley, flaxseed, and rye — along with livestock counts.

Coverage: 1905–1929 for Alberta.

### SOURCE 02

#### Treaty Boundaries & Coded Clauses

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Spatial shapefiles demarcating the geographic extent of each Numbered Treaty. Combined with a hand-coded dataset of treaty clauses — recording exactly what agricultural provisions (implements, cattle, seed) were promised under each treaty.

Enables the RDD boundary comparison.

### SOURCE 03

#### Supporting Geographic Data

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Historical climate records and soil quality maps used as control variables in the spatial regression. These help isolate treaty effects from environmental differences that naturally affect agricultural productivity across regions.

Sourced from Statistics Canada and USDA archives.

*All three sources are linked at the district level using geographic identifiers from the DIA reports.*

## 06 | Research Team & Roles

*Three research assistants, each responsible for a distinct regional dataset*



**Leen AL Saleh**

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**Research Assistant**

Dataset: Saskatchewan Region

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- ▶ Data cleaning & EDA in Excel
  - ▶ PDF extraction and digitization
  - ▶ Spatial data handling



**Alex Dagher**

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**Research Assistant**

Dataset: Alberta Region

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- ▶ PDF parsing and structuring
  - ▶ Stata-based data merging
  - ▶ Conflict & duplicate resolution



**Asmita Mandal**

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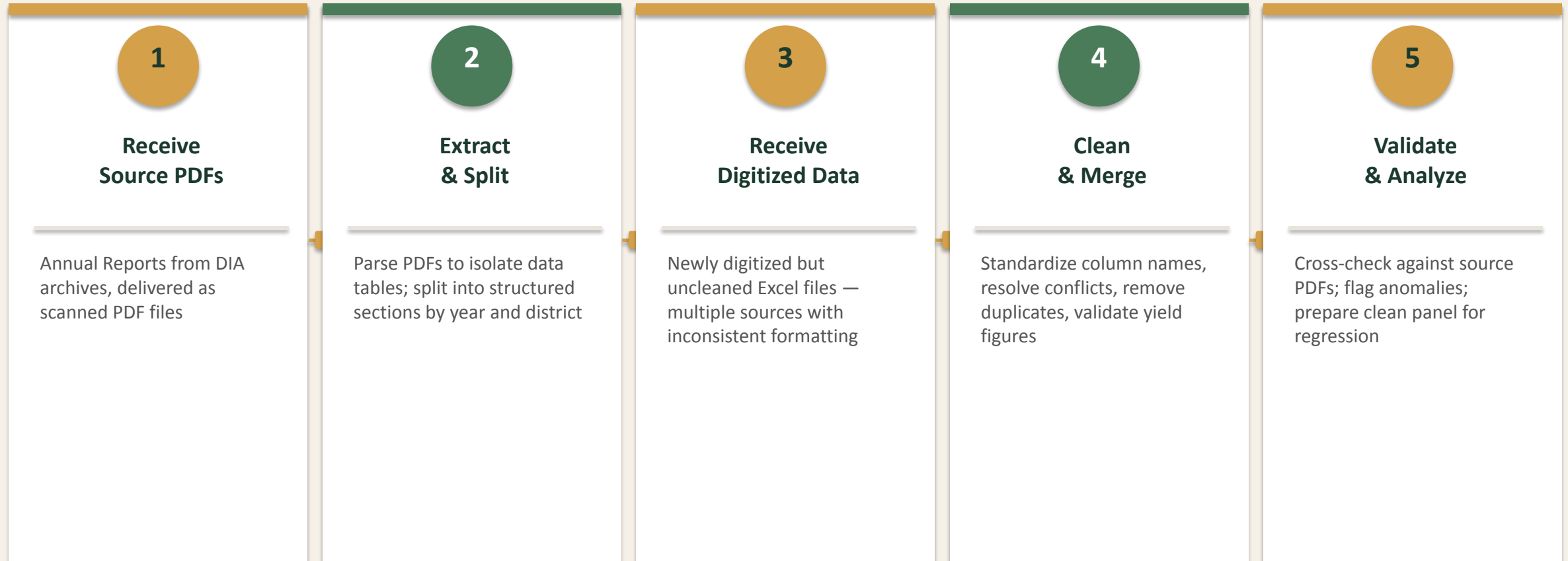
**Research Assistant**

Dataset: Manitoba Region

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- ▶ Data cleaning & validation
  - ▶ Historical document review
  - ▶ Cross-region consistency checks

Shared skills: Basic econometrics · Excel · Stata / R · Historical and spatial data handling

## 07 | Our Workflow

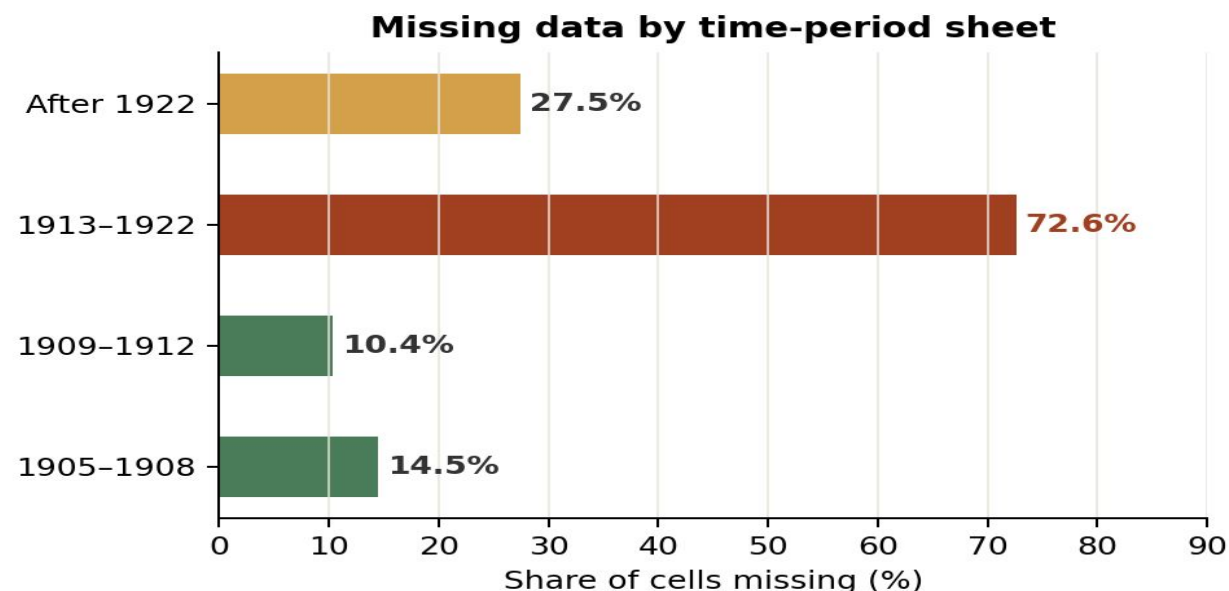


*Phase 1: Initial Setup (complete) · Phase 2: Active Cleaning (ongoing) · Phase 3: Analysis (planned)*

## 08 | Data Cleaning Challenges

### What we found

- ▶ **Column name drift**
  - ▶ Same data recorded as "Acreage" in early sheets, "Area" in later ones — breaks panel merges.
- ▶ **Extreme missing data**
  - ▶ The 1913–1922 sheet is 72.6% empty — livestock and secondary crop columns not yet digitized.
- ▶ **Implausible yield values**
  - ▶ Barley yields of 0.18 and 59 bu/acre flagged as likely OCR errors from original scans.
- ▶ **Duplicate and conflicting rows**
  - ▶ 45 records flagged across sheets, requiring cross-reference with original source PDFs.
- ▶ **Missing years (1919–1920)**
  - ▶ Two full years absent from the dataset despite being within the study window.



### How we are addressing this

- ▶ Building a column crosswalk to standardize names across all sheets
- ▶ Running automated yield cross-validation ( $\text{Area} \times \text{Yield/Acre} = \text{Total Yield}$ )
- ▶ Flagging all values outside IQR  $\pm 1.5$  for manual source-document review
- ▶ Contacting the digitization team about the 1919–1920 gap

## 09 | Progress & Next Steps

### Completed

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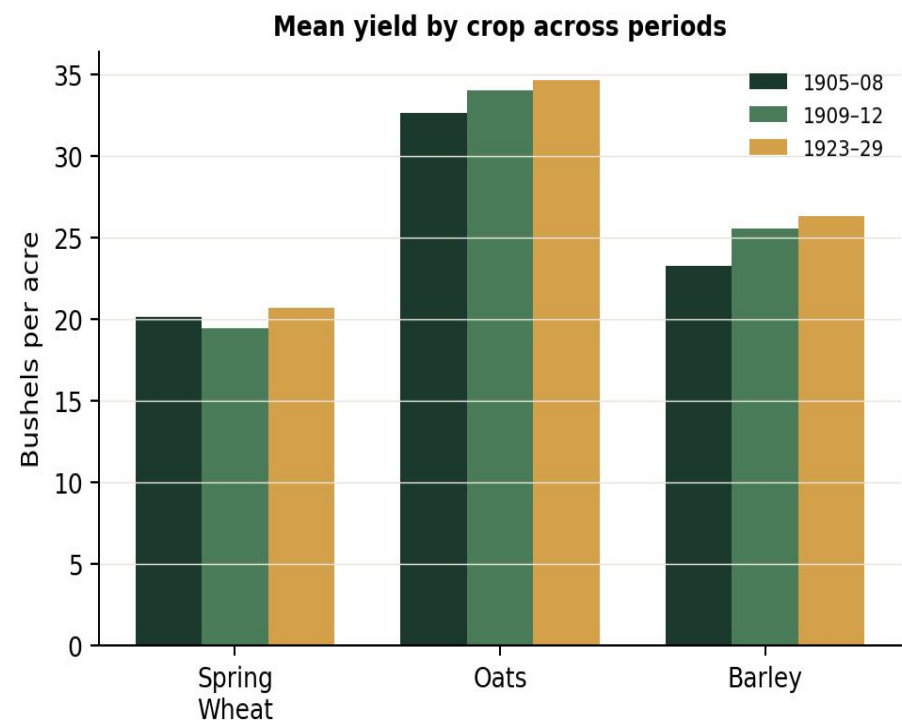
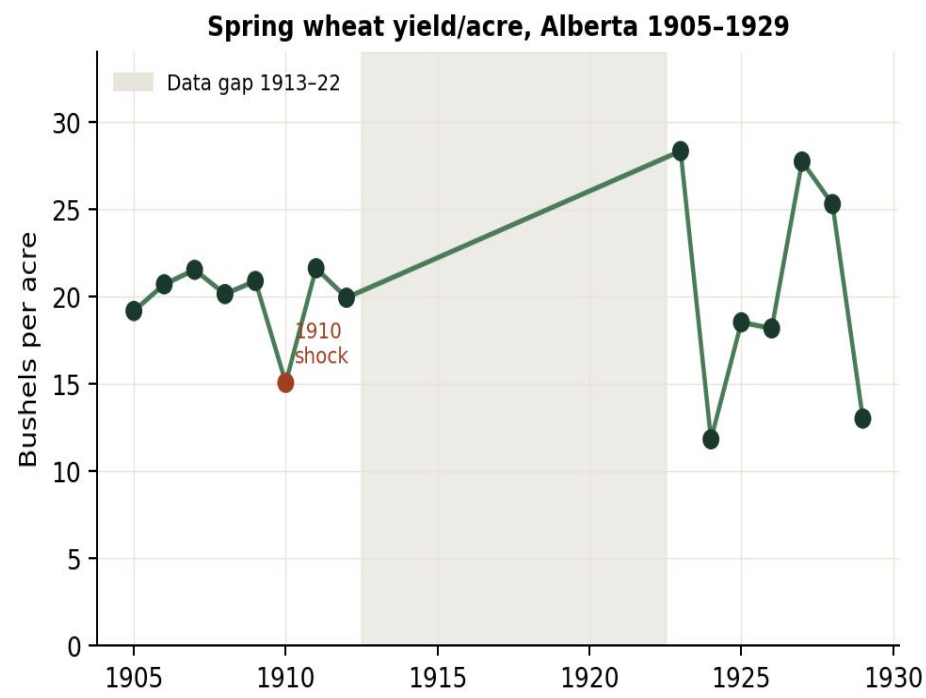
- ▶ Reviewed and extracted tables from historical DIA Annual Report PDFs
- ▶ Split source PDFs into structured sections (tables, text, appendices)
- ▶ Received digitized datasets for Alberta, Saskatchewan, and Manitoba
- ▶ Identified column schema drift and naming inconsistencies across all sheets
- ▶ Flagged 45 duplicate/conflict rows and 23 yield outliers for manual review
- ▶ Built preliminary EDA: yield trends, missing data patterns, district coverage
- ▶ Documented data quality issues in a shared Issues Log

### Next Steps

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- ▶ Complete livestock column digitization for the 1913–1922 period
- ▶ Build district crosswalk: named counties (1905–22) → numbered regions (1923+)
- ▶ Finalize clean merged panel dataset for all three regions
- ▶ Link district identifiers to treaty boundary shapefiles
- ▶ Run spatial RDD pilot — yield per acre as primary outcome
- ▶ Produce choropleth map: wheat yields overlaid on treaty boundaries
- ▶ Deliver cleaned datasets and documentation to Faculty Hisam

# 10 | Preliminary Data Insight — Alberta



## Key observations

- ▶ 1910 shows a significant crash — likely drought / weather shock
- ▶ Post-1922 yields are notably higher, suggesting mechanization gains
- ▶ Oats consistently outperform wheat in yield per acre
- ▶ Year-to-year volatility is high — climate controls are essential

*Note: These are provincial-level averages. Reserve-level data will be isolated in the next phase once treaty boundary linkage is complete.*

# Thank you

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*We welcome questions, feedback, and collaboration opportunities.*

## Research Team

Leen AL Saleh · Alex Dagher · Asmita Mandal  
Research Assistants, Department of Economics

## Faculty Supervisor

Kulsoom Hisam | Department of Economics | St. Olaf College

## Acknowledgments

We thank Faculty Hisam for guidance on this project, and acknowledge the digitization teams whose work on the DIA Annual Reports made this analysis possible.